

AI for genomics

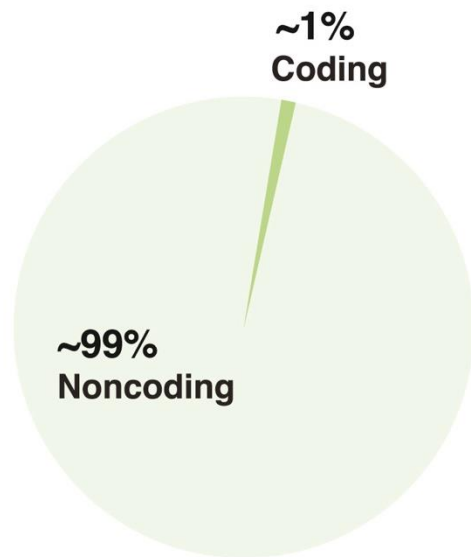
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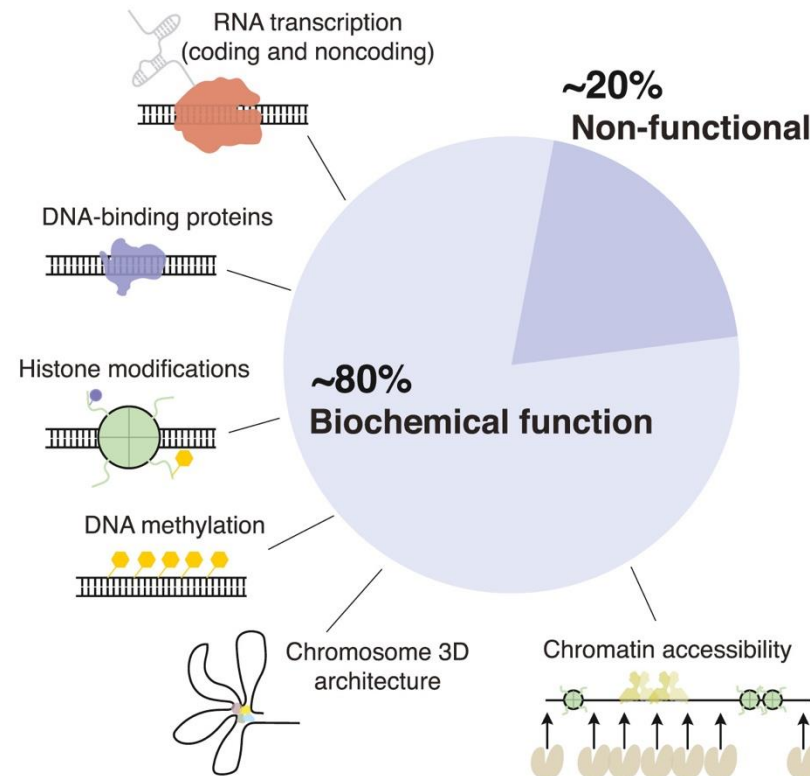
Delft Bioinformatics Lab, TUDelft

Majority of variants in non-coding regions

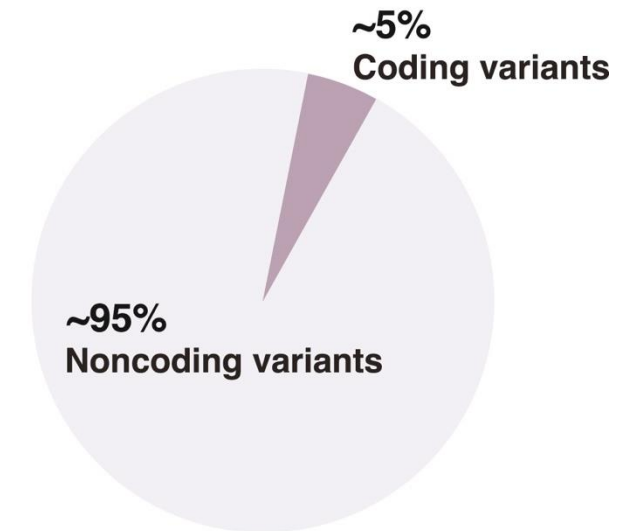
Sequence composition



Functional mapping (ENCODE)



Genome-wide association studies (GWAS)



Can we decode genome regulation rules?

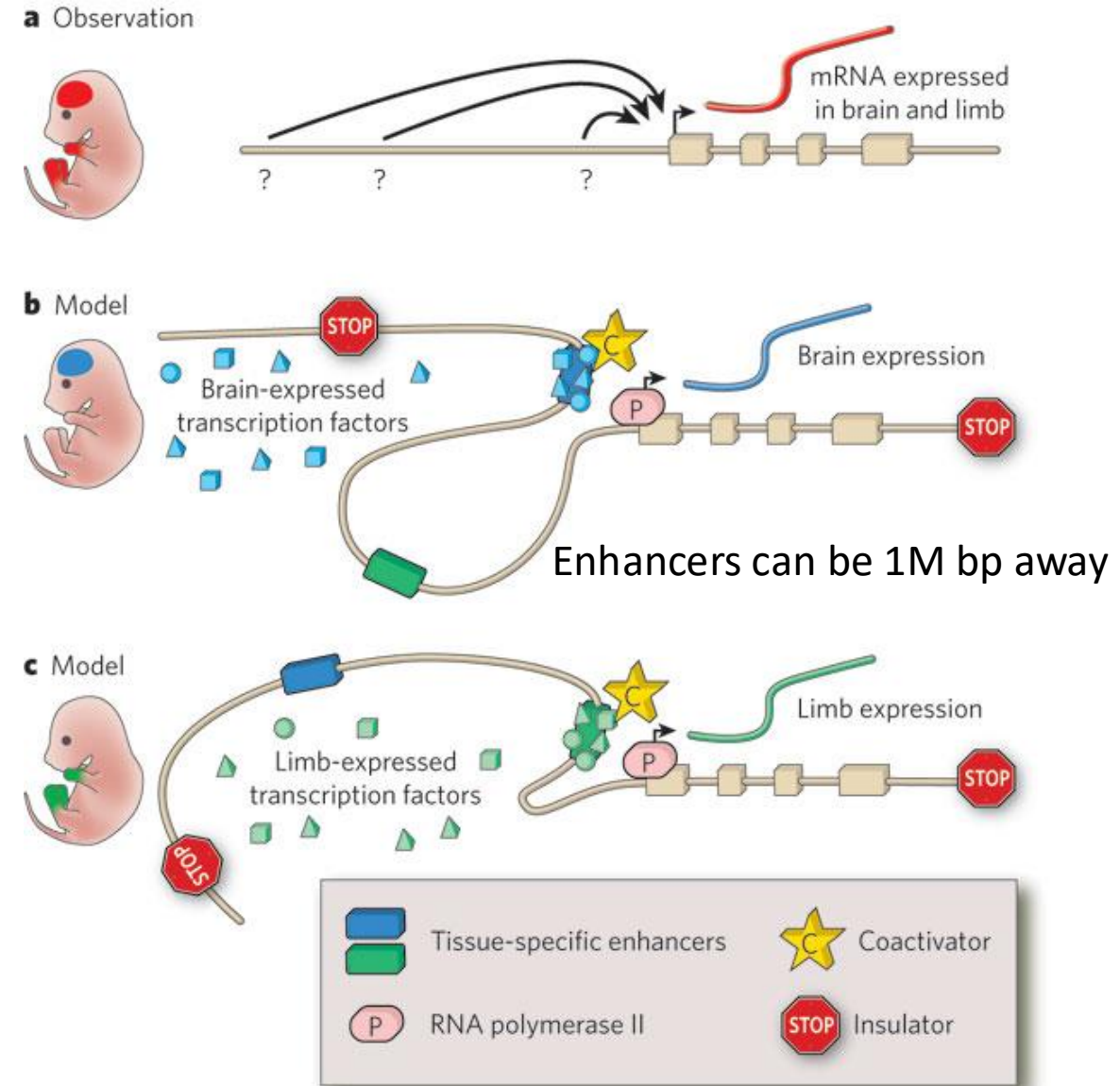
Which variant is causal in which context (cell-type)?

How are genes regulated?

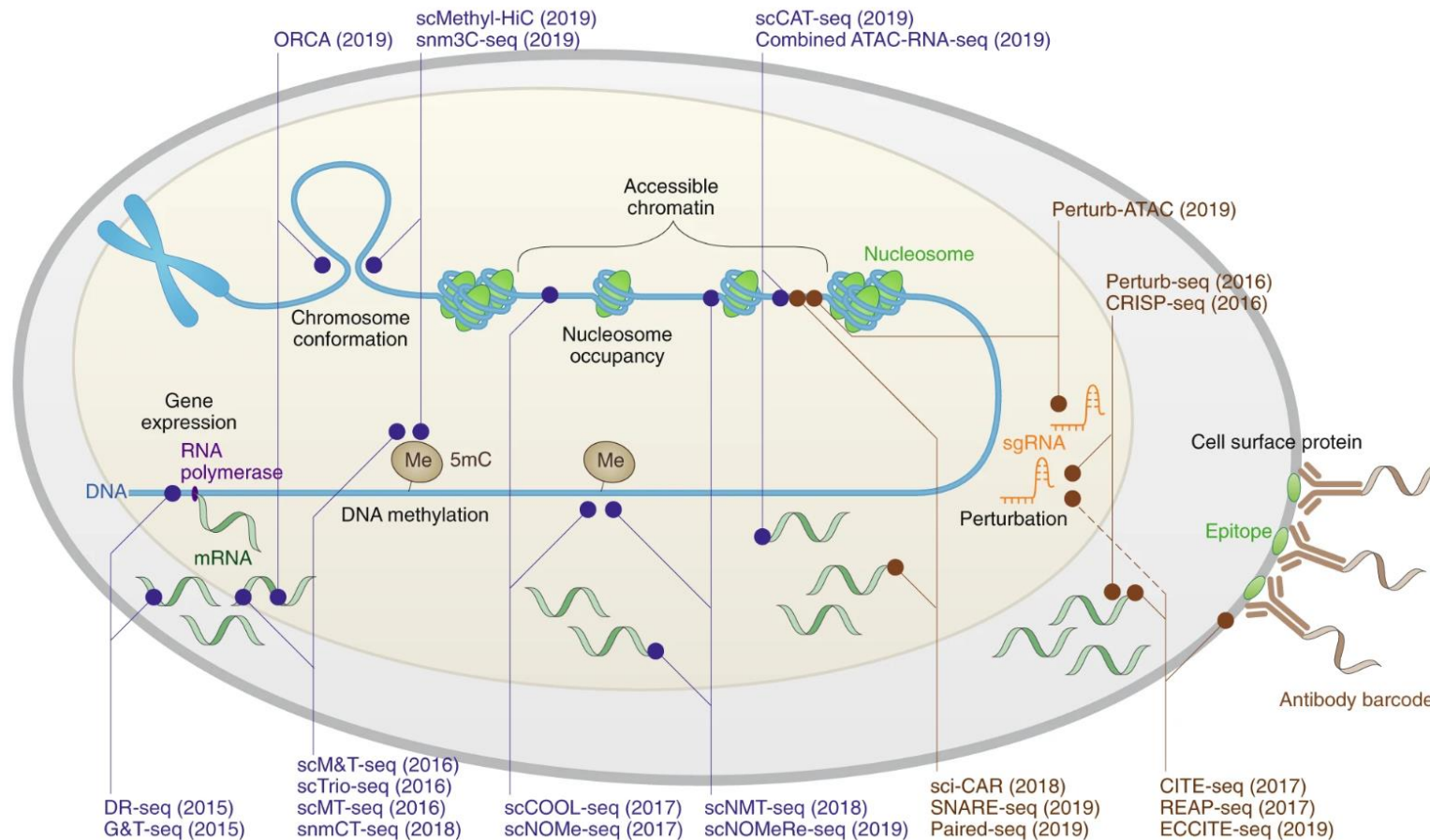
- Promoters, enhancers, TFs
- 3D chromatin organization

Other layers:

- Histone modifications
- Methylation
- RNA-splicing



Let's measure everything to figure out!

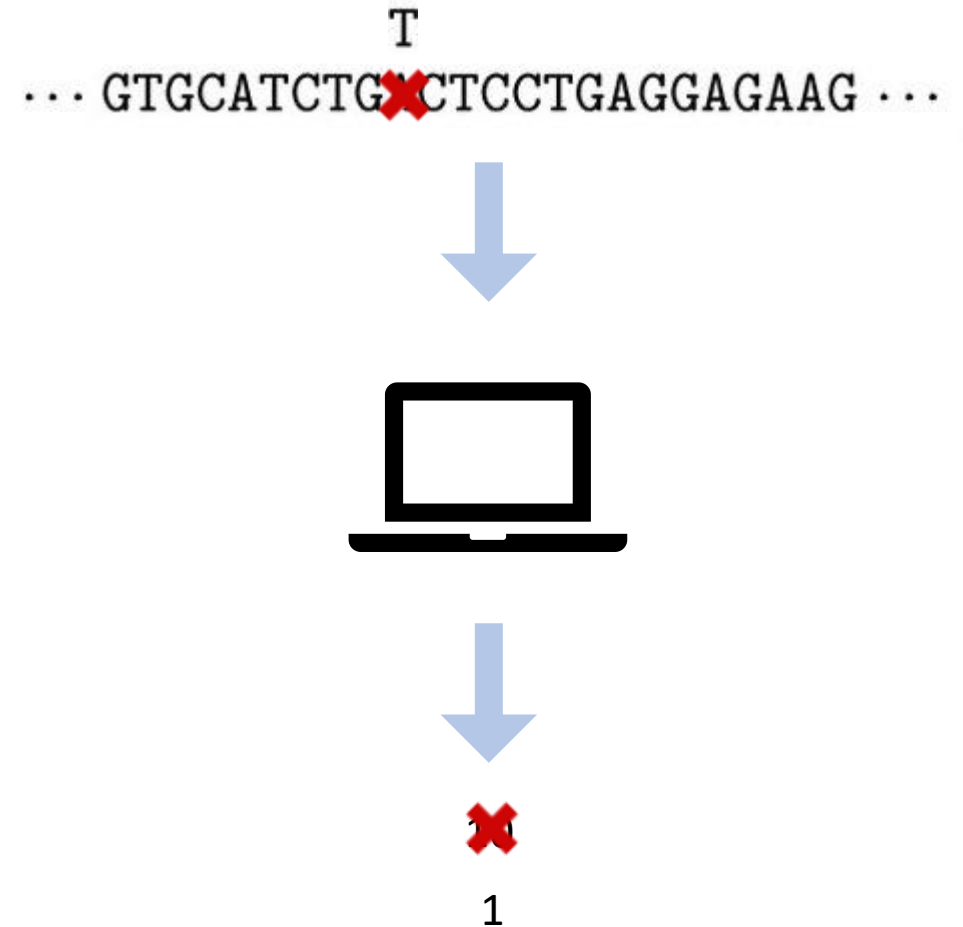


- Context specific which cell type?
- Expensive to measure multiple modalities
- Measurements can be destructive

Can we **predict** all these modalities from just the DNA sequence?

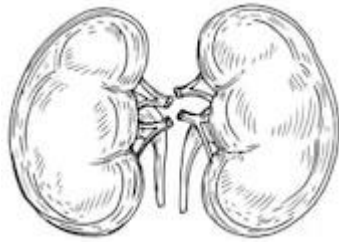
AI for genomics

- Train machine learning models to predict genomic features using the DNA sequence
 - TF binding sites
 - Chromatin accessibility
 - Histone modifications
 - Gene expression
 -



Tissue-specific predictions

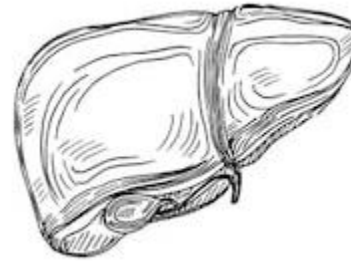
... GTGCATCTGACTCCTGAGGAGAAG ...



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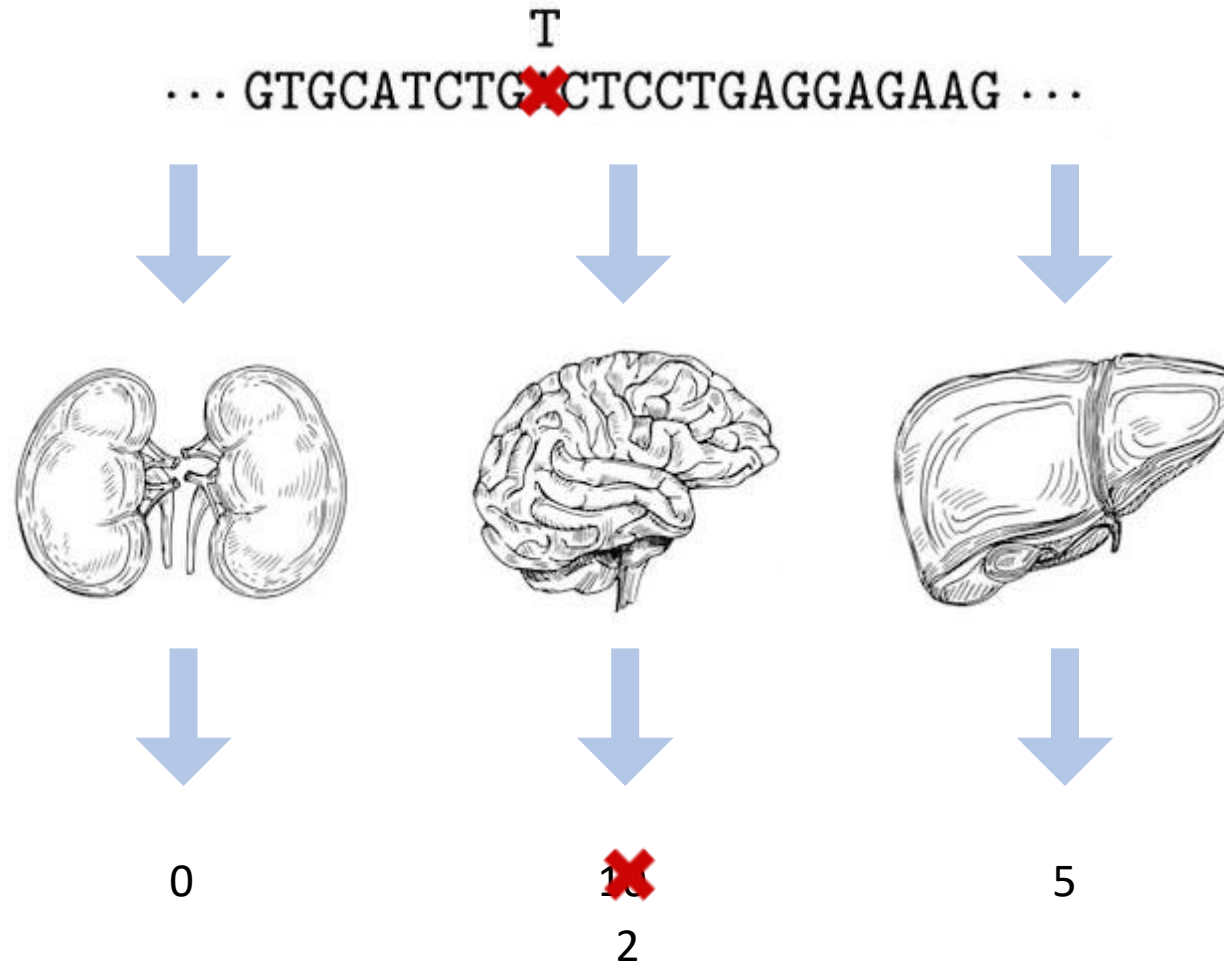


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Tissue-specific predictions



Outline

- Modeling the local sequence
 - Predicting TF binding sites
 - Predicting other genomic features
- Modeling long-range interactions
- Model interpretation

Outline

- Modeling the local sequence
 - Predicting TF binding sites
 - Predicting other genomic features
- Modeling long-range interactions
- Model interpretation

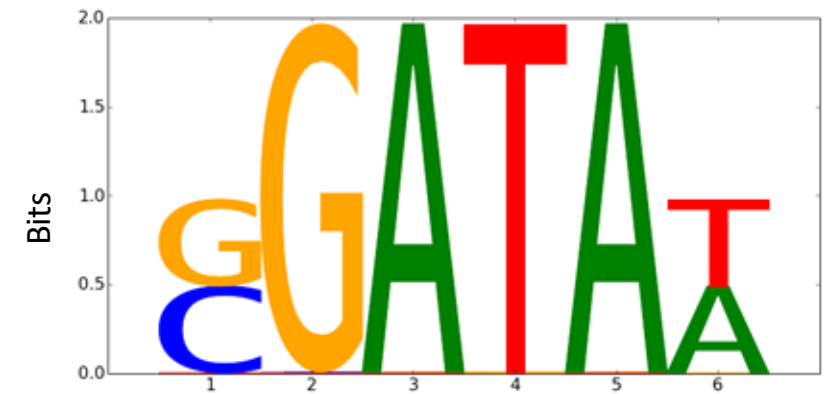
Sequence motifs

GGATAA
CGATAA
CGATAT
GGATAT

Set of aligned sequences
bound by TF

A	0	0	1	0	1	0.5
C	0.5	0	0	0	0	0
G	0.5	1	0	0	0	0
T	0	0	0	1	0	0.5

Position weight matrix
(PWM)



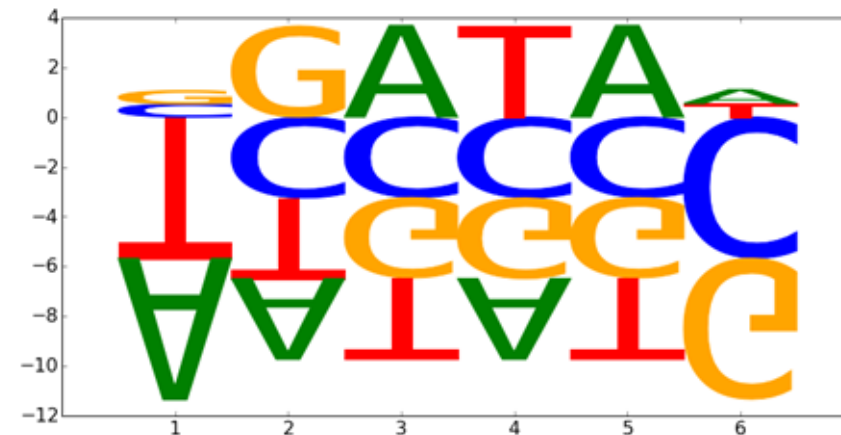
Sequence logo

Sequence motifs

- Position-specific scoring matrix (PSSM): accounting for genomic background nucleotide distribution

A	-5.7	-3.2	3.7	-3.2	3.7	0.5
C	0.5	-3.2	-3.2	-3.2	-3.2	-5.7
G	0.5	3.7	-3.2	-3.2	-3.2	-5.7
T	-5.7	-3.2	-3.2	3.7	-3.2	0.5

$$\log_2 \frac{p_i(x_i = a_i)}{p_{bg}(x_i = a_i)}$$



PSSMlogo

Convolution: scoring a sequence with a PSSM

- Given a particular sequence, does it contain a particular motif?

Convolution: scoring a sequence with a PSSM

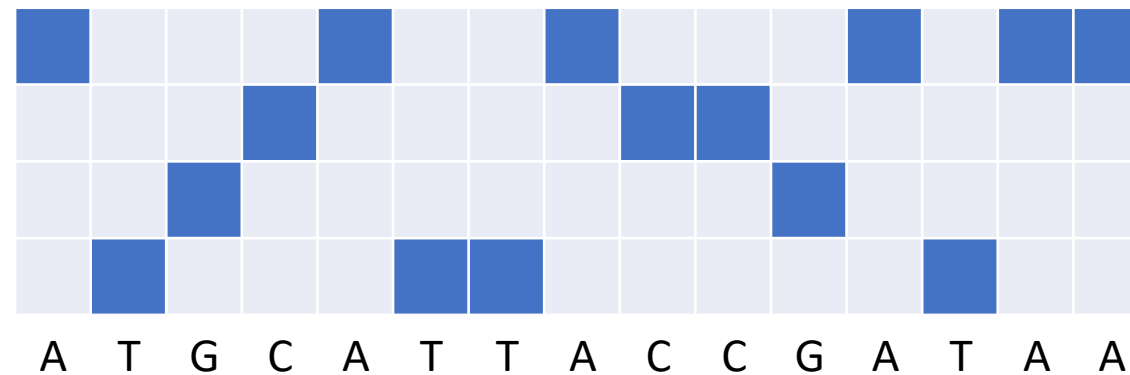
Input sequence

A T G C A T T A C C G A T A A

Convolution: scoring a sequence with a PSSM

One-hot encoding
(X)

Input sequence



Convolution: scoring a sequence with a PSSM

Motif match scores
($W * x$)

--	--	--	--	--	--	--	--	--	--

Scoring weights
(W)

A	-5.7	-3.2	3.7	-3.2	3.7	0.5
C	0.5	-3.2	-3.2	-3.2	-3.2	-5.7
G	0.5	3.7	-3.2	-3.2	-3.2	-5.7
T	-5.7	-3.2	-3.2	3.7	-3.2	0.5

One-hot encoding
(X)

■				■			■				■		■	■
			■					■	■					
		■								■				
	■				■	■						■		

Input sequence

A T G C A T T A C C G A T A A

Convolution: scoring a sequence with a PSSM

Motif match scores
($W * x$)

-11.1									
-------	--	--	--	--	--	--	--	--	--

Scoring weights
(W)

A	-5.7	-3.2	3.7	-3.2	3.7	0.5
C	0.5	-3.2	-3.2	-3.2	-3.2	-5.7
G	0.5	3.7	-3.2	-3.2	-3.2	-5.7
T	-5.7	-3.2	-3.2	3.7	-3.2	0.5

$$\begin{aligned}
 & -5.7 * 1 + 0.5 * 0 + 0.5 * 0 - 5.7 * 0 + \\
 & -3.2 * 0 - 3.2 * 0 + 3.7 * 0 - 3.2 * 1 + \\
 & \dots \\
 & 0.5 * 0 - 5.7 * 0 - 5.7 * 0 + 0.5 * 1 = -11.1
 \end{aligned}$$

One-hot encoding
(X)

Input sequence

A T G C A T T A C C G A T A A

Convolution: scoring a sequence with a PSSM

Motif match scores
($W * x$)

-11.1	-11.1								
-------	-------	--	--	--	--	--	--	--	--

Scoring weights
(W)

A	-5.7	-3.2	3.7	-3.2	3.7	0.5
C	0.5	-3.2	-3.2	-3.2	-3.2	-5.7
G	0.5	3.7	-3.2	-3.2	-3.2	-5.7
T	-5.7	-3.2	-3.2	3.7	-3.2	0.5

One-hot encoding
(X)

Input sequence

A T G C A T T A C C G A T A A

Convolution: scoring a sequence with a PSSM

Motif match scores
($W * x$)

-11.1	-11.1	2.0							
-------	-------	-----	--	--	--	--	--	--	--

Scoring weights
(W)

A	-5.7	-3.2	3.7	-3.2	3.7	0.5
C	0.5	-3.2	-3.2	-3.2	-3.2	-5.7
G	0.5	3.7	-3.2	-3.2	-3.2	-5.7
T	-5.7	-3.2	-3.2	3.7	-3.2	0.5

One-hot encoding
(X)

Input sequence

A T G C A T T A C C G A T A A

Convolution: scoring a sequence with a PSSM

Motif match scores
($W * x$)

-11.1	-11.1	2.0	-4.2						
-------	-------	-----	------	--	--	--	--	--	--

Scoring weights
(W)

A	-5.7	-3.2	3.7	-3.2	3.7	0.5
C	0.5	-3.2	-3.2	-3.2	-3.2	-5.7
G	0.5	3.7	-3.2	-3.2	-3.2	-5.7
T	-5.7	-3.2	-3.2	3.7	-3.2	0.5

One-hot encoding
(X)

■	□	□	□	■	□	□	■	□	□	□	■	□	■	■
□	□	□	■	□	□	□	□	■	■	□	□	□	□	□
□	□	■	□	□	□	□	□	□	□	■	□	□	□	□
□	■	□	□	□	■	■	□	□	□	□	□	■	□	□

Input sequence

A T G C A T T A C C G A T A A

Convolution: scoring a sequence with a PSSM

Motif match scores
($W * x$)

-11.1	-11.1	2.0	-4.2	-24.2	-17.3	-18.0	-11.1	-11.8	15.8
-------	-------	-----	------	-------	-------	-------	-------	-------	------

Scoring weights
(W)

A	-5.7	-3.2	3.7	-3.2	3.7	0.5
C	0.5	-3.2	-3.2	-3.2	-3.2	-5.7
G	0.5	3.7	-3.2	-3.2	-3.2	-5.7
T	-5.7	-3.2	-3.2	3.7	-3.2	0.5

One-hot encoding
(X)

■				■			■				■		■	■
			■					■	■					
		■								■				
	■				■	■						■		

Input sequence

A T G C A T T A C C G A T A A

Convolution: scoring a sequence with a PSSM

Thresholding the scores

0	0	2.0	0	0	0	0	0	0	15.9
---	---	-----	---	---	---	---	---	---	------

Motif match scores
($W * x$)

-11.1	-11.1	2.0	-4.2	-24.2	-17.3	-18.0	-11.1	-11.8	15.8
-------	-------	-----	------	-------	-------	-------	-------	-------	------

Scoring weights
(W)

A	-5.7	-3.2	3.7	-3.2	3.7	0.5
C	0.5	-3.2	-3.2	-3.2	-3.2	-5.7
G	0.5	3.7	-3.2	-3.2	-3.2	-5.7
T	-5.7	-3.2	-3.2	3.7	-3.2	0.5

One-hot encoding
(X)

A	T	G	C	A	T	T	A	C	C	G	A	T	A	A

Input sequence

What if the PSSM is unknown?

Measured binding sites (targets)

0	0	1	0	0	0	0	0	0	1
---	---	---	---	---	---	---	---	---	---

Motif match scores
($W * x$)

--	--	--	--	--	--	--	--	--	--

Scoring weights
(W)

A	w_1	w_2	w_3	w_4	w_5	w_6
C	w_7	w_8	w_9	w_{10}	w_{11}	w_{12}
G	w_{13}	w_{14}	w_{15}	w_{16}	w_{17}	w_{18}
T	w_{19}	w_{20}	w_{21}	w_{22}	w_{23}	w_{24}

One-hot encoding
(X)

Input sequence

A T G C A T T A C C G A T A A

What if the PSSM is unknown?

Measured binding sites (targets)

0	0	1	0	0	0	0	0	0	1
---	---	---	---	---	---	---	---	---	---

Motif match scores
($W * x$)

--	--	--	--	--	--	--	--	--	--

Scoring weights
(W)

A	-1.5	-1.4	-1.3	1.3	0.5	-0.6
C	1.1	-0.8	0.7	-0.3	0.4	-0.9
G	-1.0	-0.1	0.0	1.5	-1.4	1.1
T	1.0	-0.5	-0.9	-1.1	-1.0	-0.3

1. Randomly initialize W

One-hot encoding
(X)

1	0	0	0	1	0	0	1	0	0	0	1	0	1	1
0	0	0	1	0	0	0	0	1	1	0	0	0	0	0
0	0	1	0	0	0	0	0	0	0	1	0	0	0	0
0	1	0	0	0	1	1	0	0	0	0	0	1	0	0

Input sequence

A T G C A T T A C C G A T A A

What if the PSSM is unknown?

Measured binding sites (targets)

0	0	1	0	0	0	0	0	0	1
---	---	---	---	---	---	---	---	---	---

Motif match scores
($W * x$)

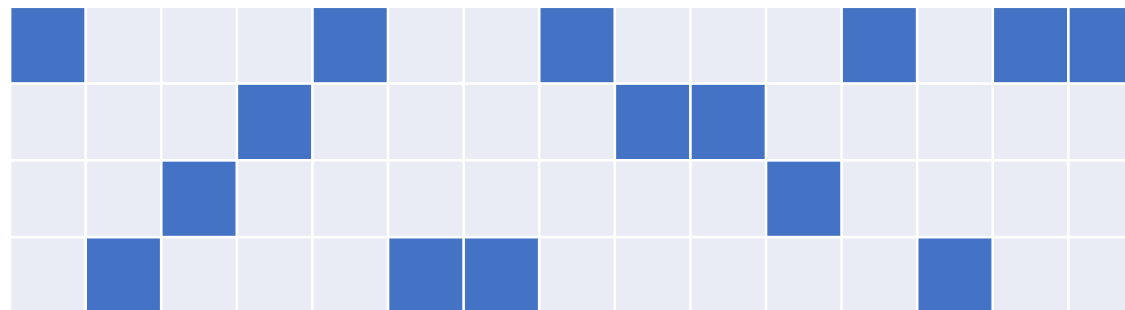
-2.1	1.6	-5.8	-2.7	-2.1	0.4	-2.0	0.1	0.0	-1.5
------	-----	------	------	------	-----	------	-----	-----	------

Scoring weights
(W)

A	-1.5	-1.4	-1.3	1.3	0.5	-0.6
C	1.1	-0.8	0.7	-0.3	0.4	-0.9
G	-1.0	-0.1	0.0	1.5	-1.4	1.1
T	1.0	-0.5	-0.9	-1.1	-1.0	-0.3

1. Randomly initialize W
2. Convolution between input sequences and scoring weights

One-hot encoding
(X)



Input sequence

A T G C A T T A C C G A T A A

What if the PSSM is unknown?

Measured binding sites (targets)

0	0	1	0	0	0	0	0	0	1
---	---	---	---	---	---	---	---	---	---

Motif match scores
($W * x$)

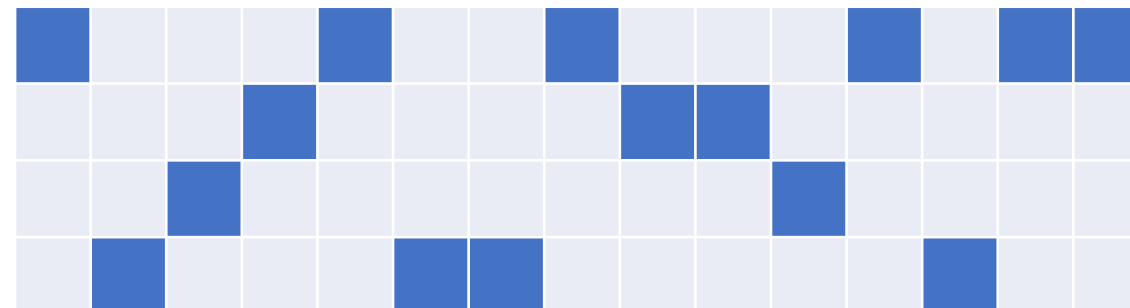
-2.1	1.6	-5.8	-2.7	-2.1	0.4	-2.0	0.1	0.0	-1.5
------	-----	------	------	------	-----	------	-----	-----	------

Scoring weights
(W)

A	-1.5	-1.4	-1.3	1.3	0.5	-0.6
C	1.1	-0.8	0.7	-0.3	0.4	-0.9
G	-1.0	-0.1	0.0	1.5	-1.4	1.1
T	1.0	-0.5	-0.9	-1.1	-1.0	-0.3

1. Randomly initialize W
2. Convolution between input sequences and scoring weights
3. Activation function

One-hot encoding
(X)

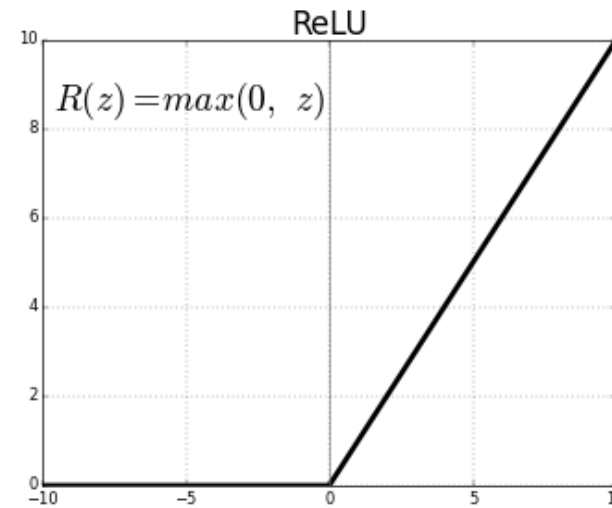
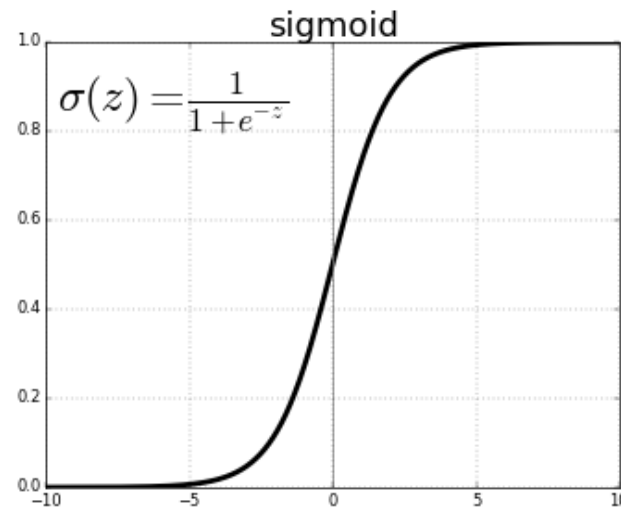


Input sequence

A T G C A T T A C C G A T A A

Activation function

- Motif match scores can be very high positive and negative numbers
- Targets are binarized (binding yes/no) or positive (how often binding was measured)
- Activation function maps the scores to the correct range



What if the PSSM is unknown?

Measured binding sites (targets)

0	0	1	0	0	0	0	0	0	1
---	---	---	---	---	---	---	---	---	---

Motif match scores
($W * x$)

0.1	0.83	0.0	0.1	0.1	0.6	0.1	0.5	0.5	0.2
-----	------	-----	-----	-----	-----	-----	-----	-----	-----

Scoring weights
(W)

A	-1.5	-1.4	-1.3	1.3	0.5	-0.6
C	1.1	-0.8	0.7	-0.3	0.4	-0.9
G	-1.0	-0.1	0.0	1.5	-1.4	1.1
T	1.0	-0.5	-0.9	-1.1	-1.0	-0.3

1. Randomly initialize W
2. Convolution between input sequences and scoring weights
3. Activation function

One-hot encoding
(X)

1	0	0	0	1	0	0	1	0	0	1	0	1	1
0	0	0	1	0	0	0	0	1	1	0	0	0	0
0	0	1	0	0	0	0	0	0	0	1	0	0	0
0	1	0	0	0	1	1	0	0	0	0	1	0	0

Input sequence

A T G C A T T A C C G A T A A

What if the PSSM is unknown?

Measured binding sites (targets)

Calculate the loss

Motif match scores
($W * x$)

0	0	1	0	0	0	0	0	0	1
0.1	0.83	0.0	0.1	0.1	0.6	0.1	0.5	0.5	0.2

Scoring weights
(W)

A	-1.5	-1.4	-1.3	1.3	0.5	-0.6
C	1.1	-0.8	0.7	-0.3	0.4	-0.9
G	-1.0	-0.1	0.0	1.5	-1.4	1.1
T	1.0	-0.5	-0.9	-1.1	-1.0	-0.3

1. Randomly initialize W
2. Convolution between input sequences and scoring weights
3. Activation function
4. Compare predictions to targets

One-hot encoding
(X)

1	0	0	0	1	0	0	1	0	0	0	1	0	1	1
0	0	0	1	0	0	0	0	1	1	0	0	0	0	0
0	0	1	0	0	0	0	0	0	0	1	0	0	0	0
0	1	0	0	0	1	1	0	0	0	0	0	1	0	0

Input sequence

A T G C A T T A C C G A T A A

What if the PSSM is unknown?

Measured binding sites (targets)

0	0	1	0	0	0	0	0	0	1
---	---	---	---	---	---	---	---	---	---

Motif match scores
($W * x$)

0.1	0.6	0	0	0	0.1	0	0.3	0.3	0.7
-----	-----	---	---	---	-----	---	-----	-----	-----

Scoring weights
(W)

A	-1.6	-1.9	-1.0	0.6	0.8	-0.1
C	1.2	-1.1	0.2	-0.9	-0.1	-1.7
G	-0.9	0.5	-0.2	0.6	-1.5	0.2
T	0.5	-1.0	-1.2	-0.5	-1.4	0.1

1. Randomly initialize W
2. Convolution between input sequences and scoring weights
3. Activation function
4. Compare predictions to targets
5. Update W
6. Repeat steps 2-5 until convergence

One-hot encoding
(X)

1	0	0	0	1	0	0	1	0	0	0	1	0	1	1
0	0	0	1	0	0	0	0	1	1	0	0	0	0	0
0	0	1	0	0	0	0	0	0	0	1	0	0	0	0
0	1	0	0	0	1	1	0	0	0	0	0	1	0	0

Input sequence

A T G C A T T A C C G A T A A

What if the PSSM is unknown?

Measured binding sites (targets)

0	0	1	0	0	0	0	0	0	1
---	---	---	---	---	---	---	---	---	---

Motif match scores
($W * x$)

0	0	0.9	0	0	0	0	0	0	1
---	---	-----	---	---	---	---	---	---	---

Scoring weights
(W)

A	-5.7	-3.2	3.7	-3.2	3.7	0.5
C	0.5	-3.2	-3.2	-3.2	-3.2	-5.7
G	0.5	3.7	-3.2	-3.2	-3.2	-5.7
T	-5.7	-3.2	-3.2	3.7	-3.2	0.5

1. Randomly initialize W
2. Convolution between input sequences and scoring weights
3. Activation function
4. Compare predictions to targets
5. Update W
6. Repeat steps 2-5 until convergence

One-hot encoding
(X)

1	0	0	0	1	0	0	1	0	0	0	1	0	1	1
0	0	0	1	0	0	0	0	1	1	0	0	0	0	0
0	0	1	0	0	0	0	0	0	0	1	0	0	0	0
0	1	0	0	0	1	1	0	0	0	0	0	1	0	0

Input sequence

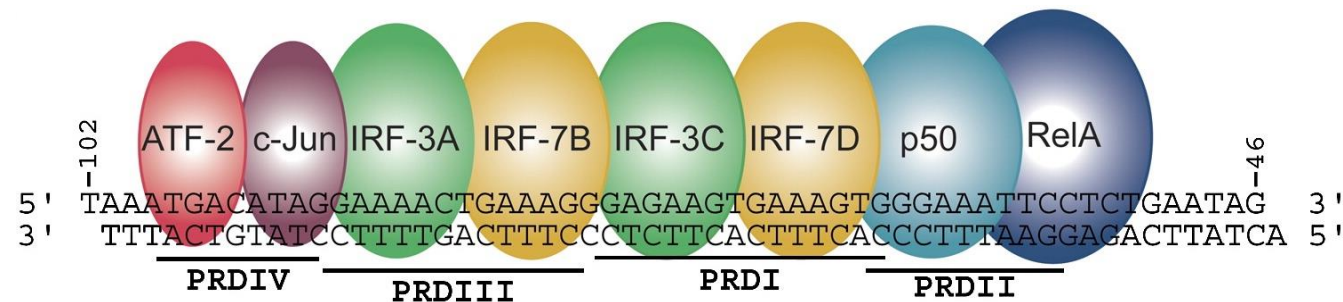
A T G C A T T A C C G A T A A

What if the PSSM is unknown?

- Example of a very small convolutional neural network (CNN)
- Train this CNN using the reference genome
- Divide reference genome in bins of ~128bp
- Divide bins into train – validation – test set
 - Train: learn the weights of the filter
 - Validation: optimize hyperparameters of the network
 - Test: evaluate the performance of your model

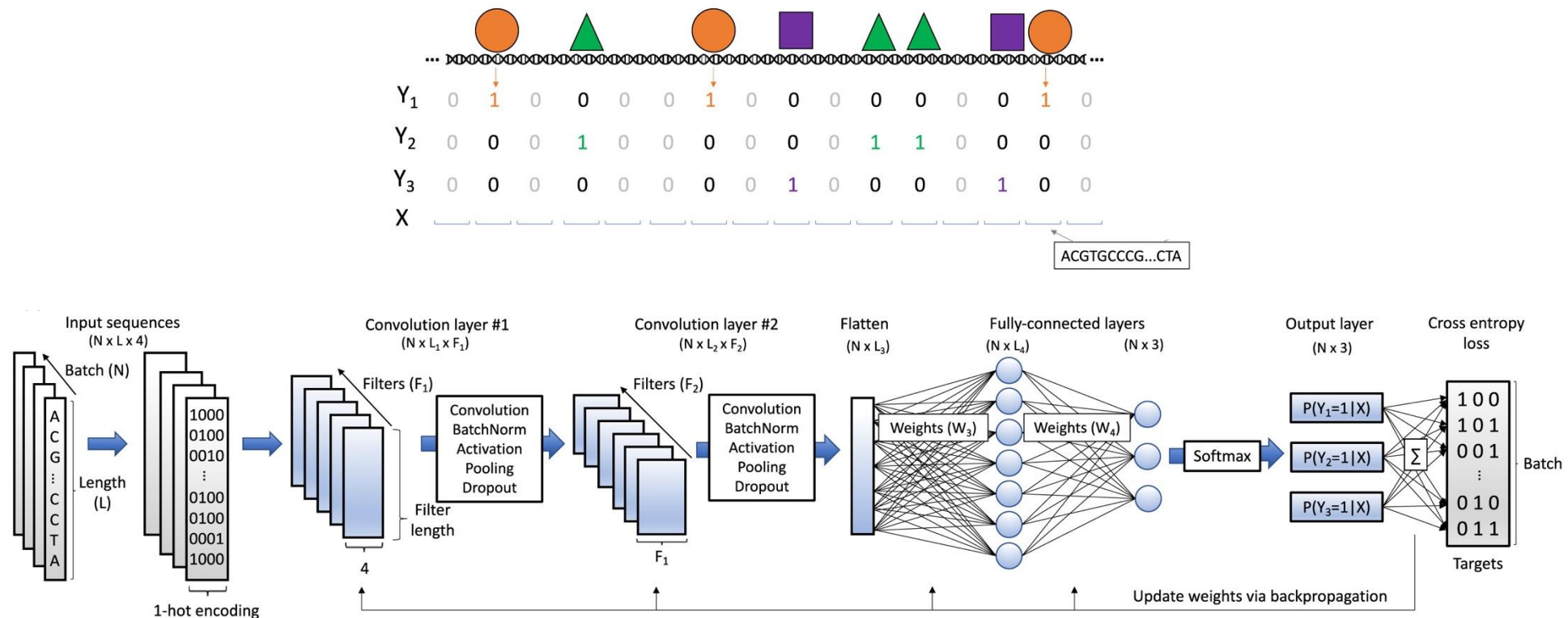
Usually, it's a bit more complex...

- TFs prefer to bind together
- Spacing between the motifs is important



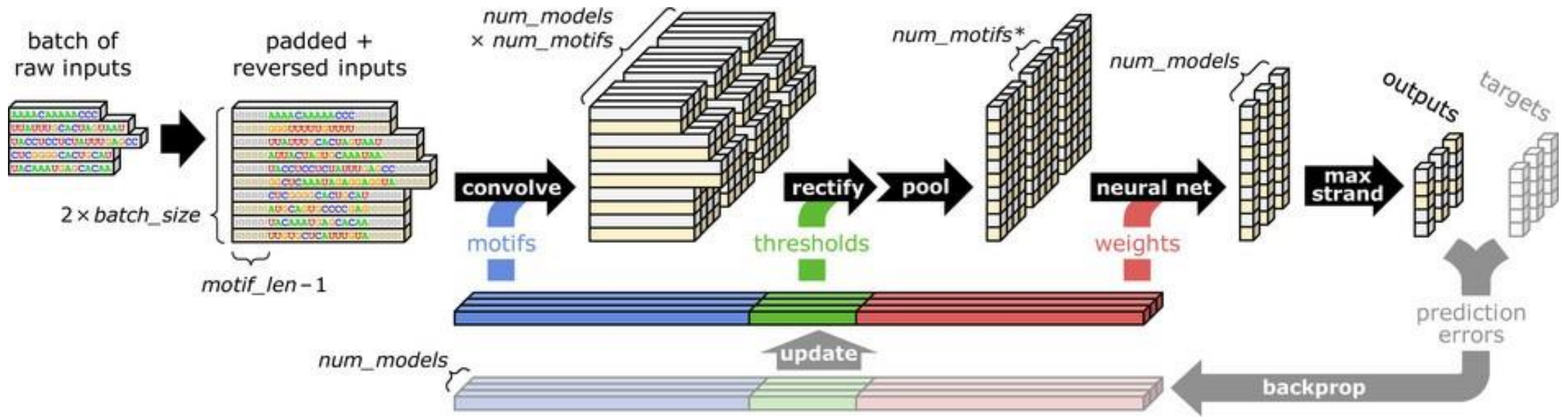
Deep convolutional neural networks

- Multitask learning: predict binding sites for multiple TF simultaneously

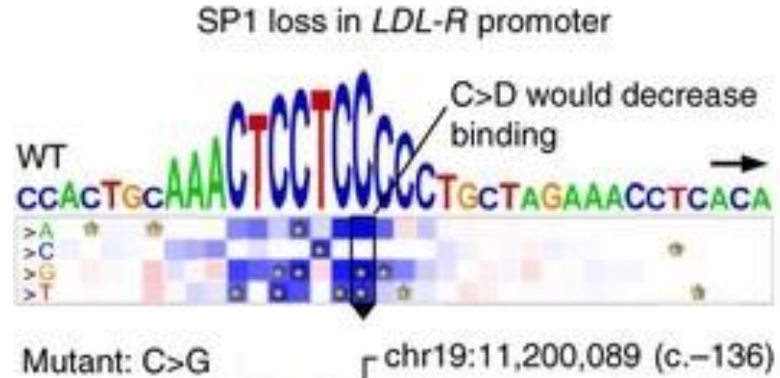


DeepBind

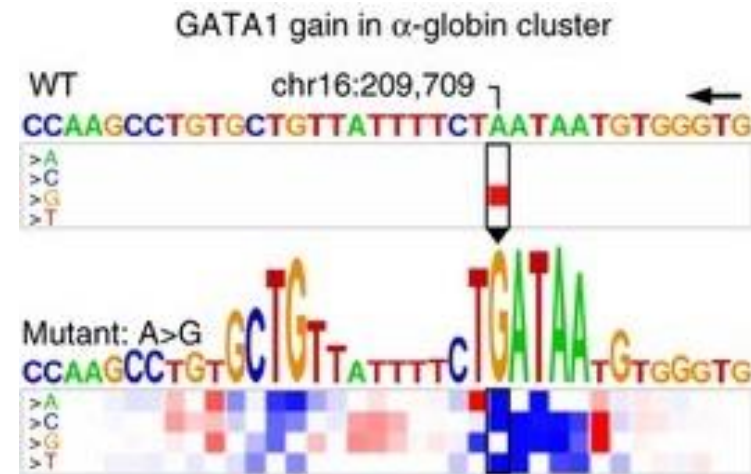
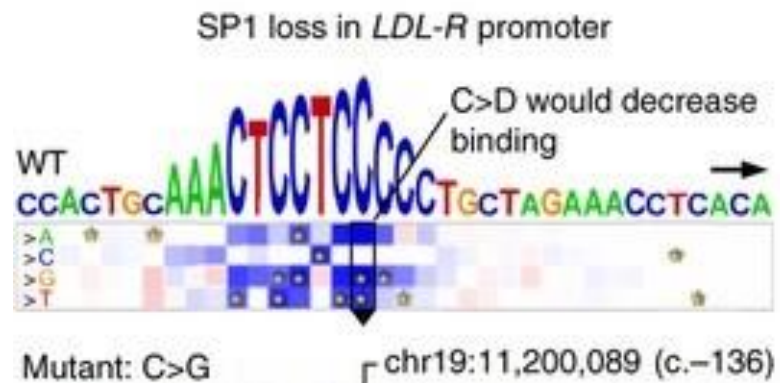
- Input: sequence of 14-101 bp
- Output: predicted binding for 538 TFs and 194 RNA binding proteins



Analysing potentially disease-causing variants



Analysing potentially disease-causing variants



Outline

- Modeling the local sequence
 - Predicting TF binding sites
 - Predicting other genomic features
- Modeling long-range interactions
- Model interpretation

Predicting other genomic features

- Is there a TF binding site in this bin?
- Is this part of the DNA accessible?
- Are there histone modifications here?

... GTGCATCTGACTCCTGAGGAGAAG ...



10

Predicting chromatin-related features

Basset: learning the regulatory code of the accessible genome with deep convolutional neural networks

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Predicting effects of noncoding variants with deep learning-based sequence model

[Jian Zhou](#) & [Olga G Troyanskaya](#) 

[Nature Methods](#) **12**, 931–934 (2015) | [Cite this article](#)

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Predicting chromatin-related features

Basset

- Input sequence: 600bp
- Bn size: 600bp
- Predict DNase-seq peaks in 164 tissues/cell types

DeepSEA

- Input sequence: 1000 bp
- Bn size: 200 bp
- Predict 919 chromatin-related features measured in different cell types
 - TF binding sites
 - DNase
 - Hstone marks

Outline

- Modeling the local sequence
 - Predicting TF binding sites
 - Predicting other genomic features
- Modeling long-range interactions
- Model interpretation

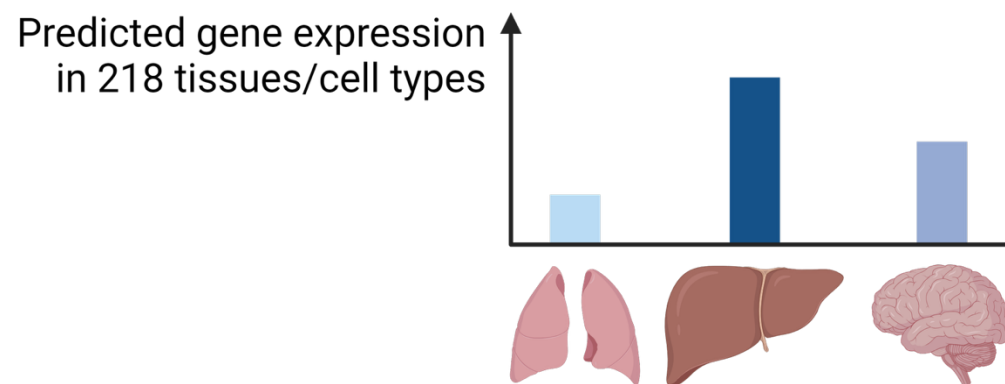
Downside of previous models

- We know the predicted effect of a mutation on a local genomic feature
- Long-range interactions are important (i.e. the DNA folds)
- What is the effect on gene expression?

ExPecto

Input sequence: 40kb, centered around TSS

... GTGCATCTGACTCCTGAGGAGAAG ...



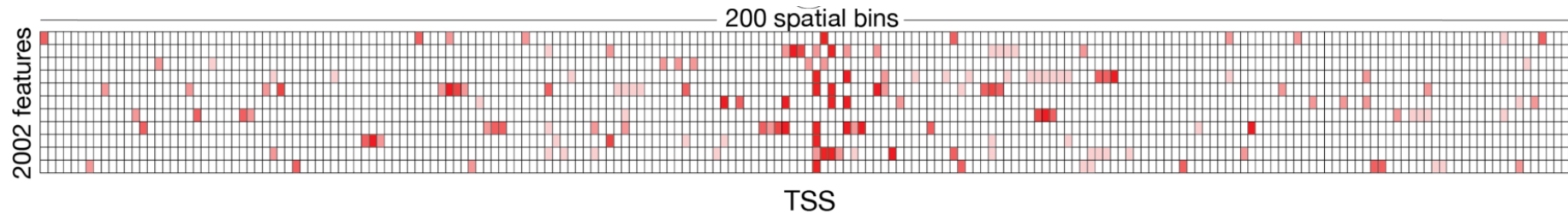
ExPecto

Input sequence: 40kb, centered around TSS

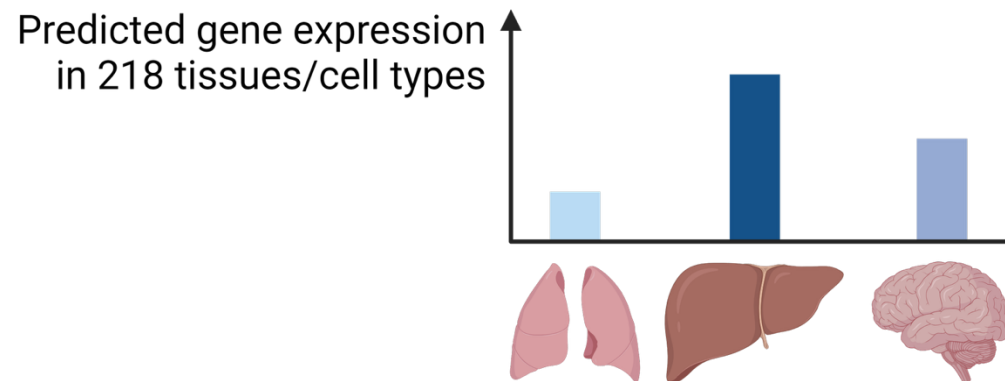
... GTGCATCTGACTCCTGAGGAGAAG ...



Use DeepSEA to predict 2002 genomic features for 200bp bins

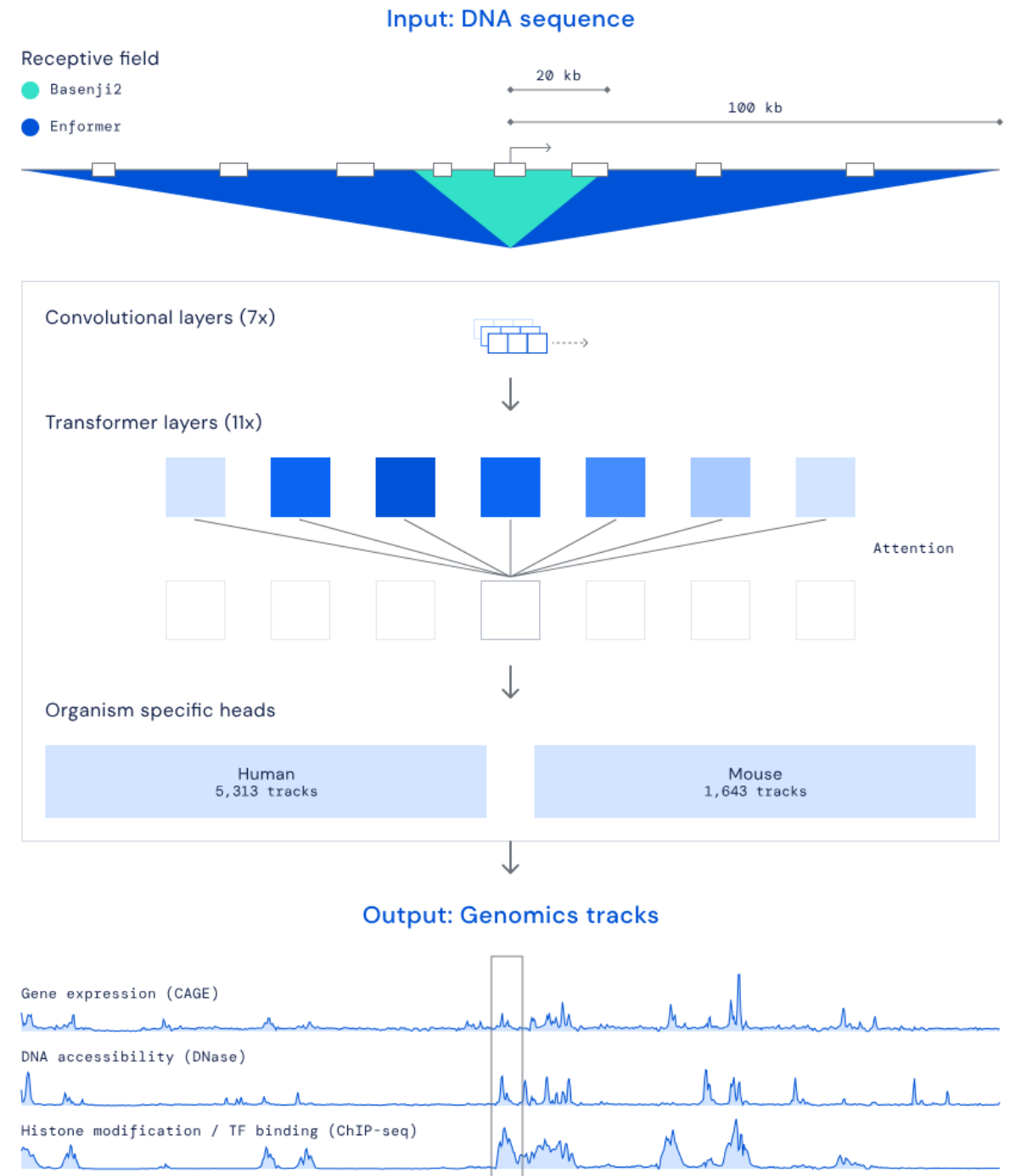


Train linear model to predict gene expression

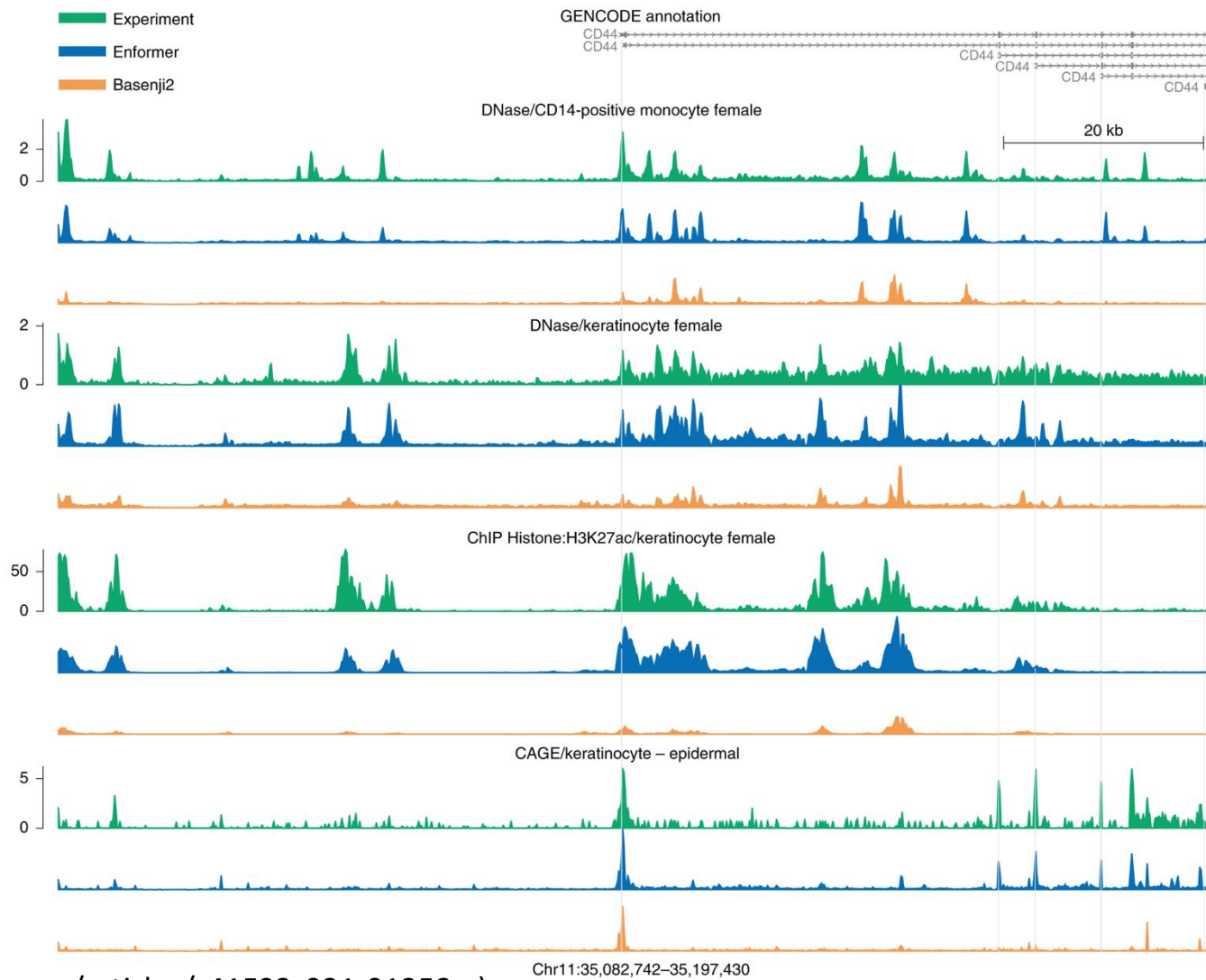


Enformer

- Predicts reads for 128bp bins
- Trained on data from human and mouse



Enformer



Outline

- Modeling the local sequence
 - Predicting TF binding sites
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In-silico saturation mutagenesis

✖TGCATCTGACT

A

C

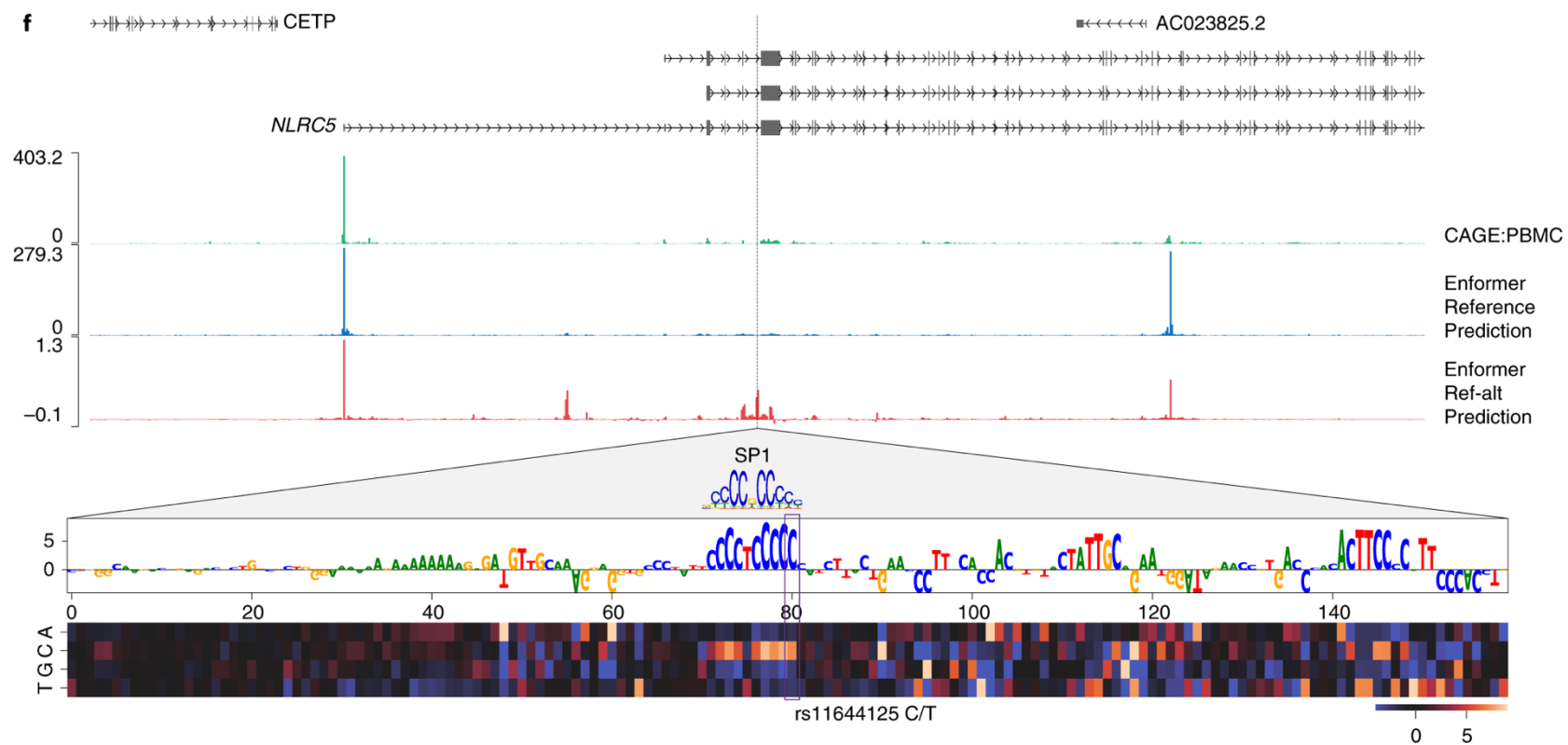
T

- ISMscore:

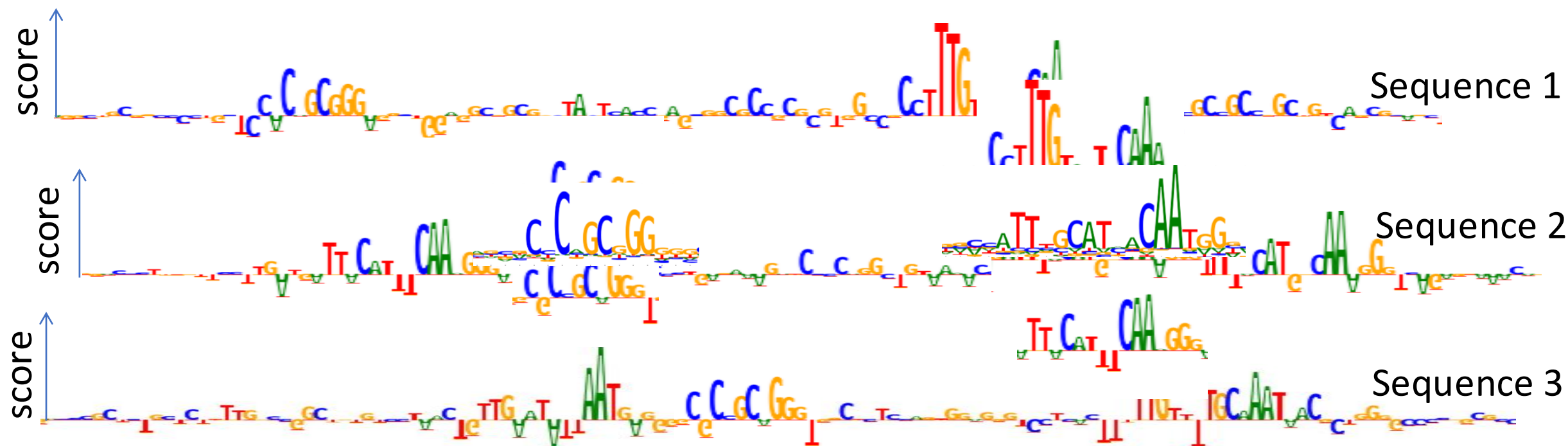
- $ISM_{g,i,n} = y_{pred,g,i,n} - \frac{1}{4} \sum_{m \in \{A,C,G,T\}} y_{pred,g,i,m}$
- Gene g
- Position i
- Nucleotide n

	G	A	C	T
y_{pred}	0.98			

In-silico saturation mutagenesis



TF-MoDISco: TF Motif Discovery from Importance Scores



Drawbacks of discussed models

- Tissues are heterogeneous → ideally we need a model for every cell type
- How to make predictions across cell types?
 - Can be solved by adding DNase as input
- Personalized predictions
 - All models are trained on the reference genome only
- Learning the effect of distal enhancers

Summary

- Deep learning models can be used to predict genomic features directly from the DNA sequence
- Advantage of using these sequence-to-prediction models
 - Improving our general understanding of biological processes in a cell
 - Predicting the effect of (non-coding) variants
- Similar models exist for the RNA sequence (e.g. predicting alternative splicing)

Useful resources

- Course material
 - Coursera deep learning course (<https://www.coursera.org/specializations/deep-learning>)
 - Seq2expr course Stanford (<https://canvas.stanford.edu/courses/174437/files>)
- AI to genomics methods
 - DeepBind (<https://www.nature.com/articles/nbt.3300>)
 - Basset (<https://genome.cshlp.org/content/26/7/990>), basenji (<https://genome.cshlp.org/content/28/5/739>), enformer (<https://www.nature.com/articles/s41592-021-01252-x>), borzoi (<https://www.biorxiv.org/content/10.1101/2023.08.30.555582v1>)
 - DeepSEA (<https://www.nature.com/articles/nmeth.3547>), Sei (<https://www.nature.com/articles/s41588-022-01102-2>), ExPecto (<https://www.nature.com/articles/s41588-018-0160-6>), ExPectoSC (<https://www.sciencedirect.com/science/article/pii/S2667237523002242?via%3Dihub>)
 - TF-MoDisco (<https://arxiv.org/abs/1811.00416>)
 - BigRNA (<https://www.biorxiv.org/content/10.1101/2023.09.20.558508v1>)
- Benchmarking papers/discussing current limitations
 - Distal enhancers (<https://genomebiology.biomedcentral.com/articles/10.1186/s13059-023-02899-9>)
 - Personalized predictions (<https://www.nature.com/articles/s41588-023-01574-w>, <https://www.nature.com/articles/s41588-023-01524-6>)